Kamesh Kotwani

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PROFILE

Azure-certified Data Scientist with 2+ years of experience delivering machine learning and NLP solutions across **healthcare and research domains**. Experienced in **Python**, **Spark**, and cloud platforms (Azure, AWS). Proven ability to build scalable pipelines, surface insights, and collaborate across teams. Eligible to join without sponsorship.

SKILLS

- **Programming & ML**: Python, R, SQL, NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow
- **Data Engineering & MLOps**: PySpark, Airflow, MLflow, DVC, Docker, CI/CD, Git, Azure (ML Studio), AWS (EC2, S3)
- **APIs & Deployment**: FastAPI, Streamlit, Flask, Docker, GitHub Actions
- **Databases & Visualisation**: Power BI, Matplotlib, Seaborn, MongoDB, MySQL
- **Development Tools**: Jupyter Notebook, VS Code, Linux, Bash

EXPERIENCE

University of Sheffield Sheffield, UK
Research Assistant / Teaching Assistant Feb 2025 – Present

- **Applied** object detection (**YOLO**) to mobile-captured blood smear images from Ghana and Uganda to identify infected regions and classify malaria parasites, supporting early-stage diagnostic research in low-resource settings.
- **Developed** NLP models to predict peer review outcomes from academic texts using **SciBERT** and traditional algorithms, trained and evaluated on **OpenReview** data using on-premise **HPC infrastructure**.
- **Mentored** postgraduate Data Science students in practical labs using **AWS**, **PySpark**, and **Databricks**, and co-developed teaching materials that improved engagement and raised course feedback scores by **15%**.

IQVIA Bangalore, India
Big Data Engineer Jun 2022 – Aug 2023

- **Built** scalable **ETL pipelines** using **Apache Spark**, **Airflow**, and **Python** on **Azure** to process millions of patient records weekly, achieving **99.8% reliability** and supporting mission-critical healthcare data applications.
- **Automated Hadoop cluster clean-up** with **Bash** and **Airflow**, reducing manual intervention and cutting recurring job failures by **40%**, significantly improving infrastructure stability.
- **Optimised** Spark workflows by refactoring transformations and upgrading legacy code, reducing pipeline runtime from **12 to 8 hours** and cutting compute costs by **8%** based on **Azure cost metrics**.
- **Resolved** high-priority production incidents via root cause analysis and cross-functional collaboration, halving incident response time and enriching the internal knowledge base using **JIRA** and **Confluence**.

CDAC Pune, India
Machine Learning Engineer Oct 2021 – Jun 2022

- **Delivered** an interactive NLP analytics dashboard using **Django**, **SpaCy**, and **Python**, integrating named entity recognition, topic modelling, and word clouds to accelerate analysis for domain experts and reduce manual review time by over **40%**.
- **Fine-tuned** multilingual transformer models using **Hugging Face** and **PyTorch** for a chatbot developed under the **Kanthasth** project, improving response accuracy by **35%** based on internal evaluation metrics.
- **Created** a user-friendly evaluation interface in **Streamlit**, enabling **non-technical testers** to validate translation outputs without scripting, significantly **reducing manual effort** and accelerating review cycles.

EDUCATION

- **MSc in Data Science**, *University of Sheffield*, Grade: *Merit* Sep 2023 – Sep 2024
- **Diploma in Big Data Analytics**, *CDAC*, Grade: *Distinction* May 2021 – Oct 2021
- **Bachelor of Engineering in Computer Science**, *GGCT*, Grade: *Distinction* Aug 2016 – Sep 2020

PROJECTS

- **RAG-based Q&A Chatbot** (`GitHub`): Developed an **LLM-based chatbot** for PDF **document search** by generating **vector embeddings** and retrieving answers via **semantic search**. Tech: **LangChain**, **OpenAI**, **FAISS**, **Streamlit**
- **Product Categorisation API** (`GitHub`): Built a **Web API** that classifies products from **text features** and **images** with **90% accuracy** across **50+ categories**. Tech: **FastAPI**, **PyTorch**, **Transformers**, **ResNet**, **Docker**
- **Real-time Sentiment Analysis API**: Developed an end-to-end **API service** that returns nuanced sentiment labels with confidence scores, achieving an **F1-score of 0.95**. Tech: **Python**, **FastAPI**, **AWS**, **MLflow**, **Docker**
- **Leaf Disease Classification App** (MSc Dissertation): Created a **Streamlit app** that classifies cassava leaf images into disease categories with a macro-**F1 score of 0.87**. Tech: **TensorFlow**, **Keras**, **MLflow**, **DVC**

CERTIFICATIONS

Microsoft Certified: Azure Data Scientist Associate (DP-100) Dec 2024
`Verify Credential`